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Multi-camera imaging system

Abstract

Systems and methods are disclosed that combine image streams from two separate imaging units into a unified, augmented image. The systems and methods can combine the image streams based on relative spatial information of the imaging units. The spatial information can include distance, angle, and rotation of the imaging units relative to one another. In some implementations, the combined image stream can be a three-dimensional stereoscopic view.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to U.S. Provisional Patent Application No. 63/112,398, which was filed on Nov. 11, 2020, the entire contents of which are incorporated herein by reference.

BACKGROUND
(1) Minimally invasive surgery involves making small incisions into a body of a patient to insert surgical tools. For example, a surgeon may perform a laparoscopic procedure using multiple cannulas inserted through individual incisions that accommodate various surgical tools including illumination devices and imaging devices. To accomplish the insertion, trocars are used to puncture the body cavity. A trocar system often includes an obturator and a cannula. An obturator is a guide placed inside a cannula with either a sharp tip (e.g., a pointed cutting blade) or blunt tip for creating an incision for the cannula. After the cannula is inserted, the obturator is removed, leaving the cannula in place for use in inserting the surgical tools. A surgical tool combining a cannula and an imaging device in a single unit is disclosed, for example. in U.S. Pat. No. 8,834,358, the disclosure of which is herein incorporated by reference in its entirety.
SUMMARY
(2) In some implementations, the present disclosure provides an imaging system including a display device, an imaging controller, a first imaging unit, a second imaging unit, and a processor. The imaging controller can include a computer-readable data storage device storing program instructions that, when executed by the processor, control the system to perform operations including receiving a first image stream and spatial information from the first imaging unit. The operations also include receiving a second image stream from the second imaging unit. The operations can further include combining the first image stream and the second image stream based the spatial information.
(3) Additionally, in some implementations, the present disclosure provides an endoscopic imaging unit including a housing, a controller, a body, a sensor housing, and antennas. The sensor housing can include an image sensor, a LiDAR device, and a light source. Also, the controller can include spatial sensors, a transmitter/receiver, and a processor. Further, the body and the sensor housing can be configured to be inserted into a body of a subject.
(4) Further, in some implementations, the present disclosure provides method including, receiving a first image stream and first spatial information from a first imaging unit. The method can also include receiving a second image stream and second spatial information from a second imaging unit. The method can further include determining spatial relationships between the first imaging unit and the second imaging unit. Additionally, the method can include determining whether one or more of the spatial relationships exceed one or one or more thresholds. Further, the method can include generating a combined image stream using the first image stream and the second image stream, and using the first spatial information and the second spatial information.

Description

DRAWINGS

(1) FIG. **1** shows a system block diagram illustrating an example of an environment for implementing systems and processes in accordance with aspects of the present disclosure.

(2) FIG. **2** shows a system block diagram illustrating examples of imaging units in accordance with aspects of the present disclosure.

(3) FIG. **3** shows a block diagram illustrating an example of a device controller in accordance with aspects of the present disclosure.

(4) FIG. **4** shows a block diagram illustrating an example of an imaging controller for a system in accordance with aspects of the present disclosure.

(5) FIGS. **5**A and **5**B show a flow block diagram illustrating an example of a method performed by a system in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

(6) The present disclosure relates generally to imaging systems and, more particularly, to endoscopic imaging systems. Systems and methods in accordance with aspects of the present disclosure combine image streams from two separate imaging units into a unified, augmented image. In some implementations, the imaging units are endoscopic devices.

(7) Systems and methods in accordance with aspects of the present disclosure can combine image streams from separate imaging units based on relative spatial information of the imaging units. In implementations, the spatial information can include, for example, distance, angle, and rotation of the imaging units relative to one another. In some implementations, the combined image stream can be, for example, a three-dimensional (“3D”) stereoscopic view. Also, in some implementations the combined image stream can have a wider field of view than individually provided by a single one of either imaging unit. Additionally, in some implementations the system can identify and characterize structures, such as tools or tissues, in the images, from combined image stream. Moreover, in some implementations the system can remove obstructions between the respective views of the imaging units from the combined image stream. Further, in some implementations, the systems and methods can also combine images from a secondary image source, such as a computed tomography (“CT”) scanner, with the combined image stream of the imaging units.

(8) Reference will now be made in detail to specific implementations illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth to provide a thorough understanding of the disclosure. However, it will be apparent to one of ordinary skill in the art that implementations may be practiced without these specific details. In other instances, known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the implementations.

(9) FIG. **1** shows a block diagram illustrating an example of an environment **100** for implementing systems and methods in accordance with aspects of the present disclosure. In some implementations, the environment **100** can include an imaging controller **105**, a display device **107**, imaging units **111**A, **111**B, a secondary imaging system **115**, and a subject **117**. The imaging controller **105** can be a computing device connected to the display device **107**, the imaging units **111**A, **111**B, and the secondary imaging system **115** through one or more wired or wireless communication channels **123**A, **123**B, **123**C, **123**D. The communication channels **123**A, **123**B, **123**C may use various serial, parallel, video transmission protocols suitable for their respective signals such as image streams **127**A, **127**B, **131**, and **133**, and data signals, such as spatial information **129**A, **129**B.

(10) The imaging controller **105** can include hardware, software, or a combination thereof for performing operations in accordance with the present disclosure. The operations can include receiving the respective image streams **127**A, **127**B and the spatial information **129**A, **129**B from

the imaging units **111**A, **111**B. The operations can also include processing the spatial information **129**A, **129**B to determine relative positions, angles, and rotations of the imaging units **111**A, **111**B. In some implementations, the images streams **127**A, **127**B and the spatial information **129**A, **129**B can be substantially synchronous, real-time information captured by the imaging units **111**A, **111**B. In some implementations, determining the relative positions, angles, and rotations includes determining respective fields-of-view of the imaging units **111**A, **111**B. For example, the relative visual perspective can include a relative distance, angle and rotation of the imaging units' **111**A, **111**B fields-of-view.

(11) The operations of the imaging controller **105** can also include combining the image streams **127**A, **127**B into the combined image stream **133** based on the spatial information **129**A, **129**B. In some implementations, combining the image streams **127**A, **127**B includes registering and stitching together the images in the fields-of-view of the imaging units **111**A, **111**B based on the spatial information **129**A, **129**B. The combined image stream **133** can provide an enhanced display **145** of the subject **117**, who can be a surgical patient. For example, the imaging controller **105** can generate the combined image stream **133** to selectably enhance the field-of-view of one of the imaging units **111**A, **111**B based on the field-of-view of the other imaging unit **111**A, **111**B. In some implementations, the enhanced display **145** may be a stereoscopic 3D view from the perspective of one of the imaging units **111**A, **111**B.

(12) Additionally, the operations of the imaging controller **105** can include identifying, characterizing, and removing structures included in the combination of the image streams **127**A, **127**B. In some implementations, using the spatial information **129**A, **129**B, the imaging controller **105** can identify and remove images of physical structures in the overlapping fields-of-view of the imaging unit **111**A using the image streams **127**A, **127**B, and the spatial information **129**A, **129**B. For example, the obstruction can be a sensor housing (e.g., sensor housing **217**) or it can be a portion of the subject's **117** bowels in the subject's **117** body cavity that blocks the field-of-view of the first imaging unit **111**A or the second imaging unit **111**B. The imaging controller **105** can determine the size and location of the obstruction and automatically process the image streams **127**A, **127**B to remove the obstruction from the combined images stream **133** and the enhanced display **145**.

(13) The display device **107** can be one or more devices that can display the enhanced display **145** for an operator of the imaging units **111**A, **111**B. As described above, the display device **107** can receive the combined image stream **133** and display the enhanced image **145**, which can include an augmented combination of the respective image streams **127**A, **127**B from the imaging units **111**A, **111**B. The display device **107** can be a liquid crystal display (LCD) display, organic light emitting diode displays (OLED), cathode ray tube display, or other suitable display device. In some implementations, the display device **107** can be a stereoscopic head-mounted display, such as a virtual reality headset.

(14) The imaging units **111**A, **111**B can be devices including various sensors that generate the image streams **127**A, **127**B and the spatial information **129**A, **129**B. In some implementations, the imaging units **111**A, **111**B are endoscopic devices configured to be inserted in the body of the subject **117**, as previously described herein. For example, the imaging units **111**A, **111**B can be laparoscopic instruments used to visualize a body cavity of the subject **117** and record images inside the body cavity. As illustrated in FIG. **1**, the imaging units **111**A, **111**B can be inserted into the subject **117** at different locations **135**A, **135**B (e.g., ports) and positioned at an angle **137** with respect to each other so as to provide overlapping fields-of-view inside a body cavity of the subject **117**. The imaging units **111**A, **111**B can also be rotated relative to one another around their long axes. For example, the imaging controller **105** can provide feedback to a surgeon via the display device **107** to translate and rotate the imaging units **111**A, **111**B in the subject **117** such that the imaging units **111**A, **111**B have overlapping field-of-views within sufficient distance, angle, and orientation thresholds for stitching the image streams **127**A, **127**B into the combined image stream

133.

(15) The secondary imaging system **115** can be an apparatus that provides one or more alternate image streams **131** of the subject **117**. In some implementations, the alternate images **131** comprise a substantially real-time image stream. The secondary imaging system **115** can be, for example, a CT scanning system, an X-ray imaging system, an ultrasound system, a fluorescence imaging system (e.g., indocyanine green fluorescence), or other multi-spectral wavelength system. In some implementations, the imaging controller **105** can process the alternate image stream **131** to further combine them with the combination of image streams **127**A, **127**B. For example, the imaging controller **105** can generate the combination image stream **133** by overlaying a synchronized image stream with a 3D stereoscopic images generated using the image streams **127**A, **127**B.

(16) FIG. **2** shows a system diagram illustrating examples of imaging units **111**A, **111**B in accordance with aspects of the present disclosure. The imaging units **111**A, **111**B can be the same or similar to those discussed above. In some implementations, the imaging units **111**A, **111**B can include a housing **200**, a device controller **201**, a body **203**, an actuator handle **205**, a cannula **209**, an obturator **211**, a sensor housing **217**, and antennas **221**A, **221**B, **221**C. The cannula **209**, the obturator **211**, and the sensor housing **217** of the individual imaging units **111**A, **111**B can be inserted into the body of a subject (e.g., subject **117**) and positioned at an angle **137** with respect to each other so as to provide overlapping fields-of-view from the sensor housing **217**.

(17) The device controller **201** can be one or more devices that process signals and data to generate respective image streams **127**A, **127**B and spatial information **129**A, **129**B of the imaging units **111**A, **111**B. In some implementations, the device controller **201** can determine the spatial information **129**A, **129**B by processing data from spatial sensors (e.g., accelerometers) to determine the relative position, angle, and rotation of the imaging units **111**A, **111**B. Also, in some implementations, the device controller **201** can also determine the spatial information **129**A, **129**B by processing range information received from sensors (e.g., image sensor **231** and LiDAR device **233**) in the sensor housing **217**. Additionally, in some implementations, the device controller **201** can process the spatial information **129**A, **129**B by processing signals received via the antennas **221**A, **221**B, **221**C to determine relative distances of the imaging units **111**A, **111**B. It is understood that, in some implementations, only one of the imaging units **111**A, **111**B provides spatial information **129**.

(18) The cannula **209** may be formed of a variety of cross-sectional shapes. For example, the cannulas **209** can have a generally round or cylindrical, ellipsoidal, triangular, square, rectangular, and D-shaped (in which one side is flat). In some implementations, the cannula **209** includes an internal lumen **202** that retains the obturator **211**. The obturator **211** can be retractable and/or removable from the cannula **209**. In some implementations, the obturator **211** is made of solid, non-transparent material. In another implementation, all or parts of the obturator **211** are made of optically transparent or transmissive material such that the obturator **211** does not obstruct the view through the camera (discussed below).

(19) The sensor housing **217** can be integral with the cannula **209** or it may be formed as a separate component that is coupled to the cannula **209**. In either case, the sensor housing **217** can disposed on or coupled to the cannula **209** at a position proximal to the distal end of the cannula **209**. In some implementations, the sensor housing **217** can be actuated by the actuator handle **205** to open, for example, after inserted into the subject's **117** body cavity. The sensor housing **217** can reside along tube **110** in the distal direction such that it is positioned within the body cavity of a subject (e.g., subject **117**). At the same time, sensor housing **217** can be positioned proximal to distal end such that it does not interfere with the insertion of the distal end of the cannula **209** as it is inserted into a subject (e.g., subject **117**). In addition, the sensor housing **217** can positioned proximally from distal end to protect the electronic components therein as distal end creates is inserted into the subject.

(20) In some implementations, the sensor housing **217** can include one or more image sensors **231**,

a LiDAR device **233**, and a light source **235**. The light source **235** can be dimmable light-emitting device, such as a LED, a halogen bulb, an incandescent bulb, or other suitable light emitter. The image sensors **231** can be devices configured to detect light reflected from the light source **235** and output an image signal. The image sensor **231** can be, for example, a charged coupled device ("CCD") or other suitable imaging sensor. In some implementations, the image sensor **231** includes at least two lenses providing stereo imaging. In some implementations, the image sensor **231** can be an omnidirectional camera.

(21) The LiDAR device **233** can include one or more devices that illuminate a region with light beams, such as lasers, and determine distance by measuring reflected light with a photosensor. The distance can be determined based a time difference between the transmission of the beam and detection of backscattered light. For example, using the LiDAR device **233**, the device controller **201** can determine spatial information **129** by sensing the relative distance and rotation of the cannulas **209** or the sensor housing **217** inside a body cavity.

(22) Additionally, the antennas **221**A, **221**B, **221**C can be disposed along the long axis of the imaging units **111**A, **111**B. In some implementations, the antennas **221**A, **221**B, **221**C can be placed in a substantially straight line on one or more sides of the imaging units **111**A, **111**B. For example, two or more lines of the antennas **221**A, **221**B, **221**C can be located on opposing sides of the housing **203** and the cannula **209**. Although FIG. **2** shows a single line of the antennas **221**A, **221**B, **221**C on one side of the imaging units **111**A, **111**B, it is understood that the additional lines of the antennas **221**A, **221**B, **221**C can be placed in opposing halves, thirds, or quadrants of the imaging units **111**A, **111**B.

(23) As illustrated in FIG. **2**, the device controllers **201** can transmit a ranging signal **223**. In some implementations, the location signals are ultra-wideband ("UWB") radio signal usable to determine a distance between the imaging units **111**A, **111**B less than or equal to 1 centimeter based on signal phase and amplitude of the radio signals, as described in IEEE 802.15.4Z. The device controller **201** can determine the distances between the imaging units **111**A, **111**B based on the different arrival times of the ranging signals **223**A and **223**B at their respective antennas **221**A, **221**B, **221**C. For example, referring to FIG. **3**, the ranging signal **223**A emitted by imaging unit **111**A can be received by imaging unit **111**B at antenna **221**C and an amount of time (T) after arriving at antenna **221**B. By making a comparison of the varying times of arrival of the ranging signal **223**A at two or more of the antennas **221**A, **221**B, **221**C, the device controller **201** of imaging unit **111**B can determine its distance and angle from imaging unit **111**A. It is understood that the transmitters can be placed at various suitable locations within the imaging units **111**A, **111**B. For example, in some implementation, the transmitters can be located in the cannulas **209** or in the sensor housings **217**.

(24) FIG. **3** shows a functional block diagram illustrating an example of a device controller **201** in accordance with aspects of the present disclosure. The device controller **201** can be the same or similar to that described above. In some implementations, the device controller **201** can include a processor **305**, a memory device **307**, a storage device **309**, a communication interface **311**, a transmitter/receiver **313**, an image processor **315**, spatial sensors **317**, and a data bus **319**.

(25) In some implementations, the processor **305** can include one or more microprocessors, microchips, or application-specific integrated circuits. The memory device **307** can include one or more types of random-access memory (RAM), read-only memory (ROM) and cache memory employed during execution of program instructions. The processor **305** can use the data buses **319** to communicate with the memory device **307**, the storage device **309**, the communication interface **311**, the image processor **315**, and the spatial sensors **317**. The storage device **309** can comprise a computer-readable, non-volatile hardware storage device that stores information and program instructions. For example, the storage device **309** can be one or more, flash drives and/or hard disk drives. The transmitter/receiver **313** can be one or more devices that encodes/decodes data into wireless signals, such as the ranging signal **223**.

(26) The processor **305** executes program instructions (e.g., an operating system and/or application

programs), which can be stored in the memory device **307** and/or the storage device **309**. The processor **305** can also execute program instructions of a spatial processing module **355** and an image processing module **359**. The spatial processing module **335** can include program instructions that determine the spatial information **129** by combining spatial data provided from the transmitter/receiver **313** and the spatial sensors **317**. The image processing module **359** can include program instructions that, using the image signal **127** from an imaging sensor (e.g., image sensor **231**), register and stitch the images to generate the image stream **127**. The image processor **315** can be a device configured to receive an image signal **365** from an image sensor (e.g., image sensor **231**) and condition images included in the image signal **365**. In accordance with aspects of the present disclosure, conditioning the image signal **365** can include normalizing the size, exposure, and brightness of the images. Also, conditioning the image signal **365** can include removing visual artifacts and stabilizing the images to reduce blurring due to motion. Additionally, the image processing module **359** can be identify and characterize structures in the images.
(27) In some implementations, the spatial sensors **317** can include one or more of, piezoelectric sensors, mechanical sensors (e.g., a microelectronic mechanical system ("MEMS"), or other suitable sensors for detecting the location, velocity, acceleration, and rotation of the imaging units (e.g., imaging units **111**A, **111**B).
(28) It is noted that the device controller **201** is only representative of various possible equivalent-computing devices that can perform the processes and functions described herein. To this extent, in some implementations, the functionality provided by the device controller **201** can be any combination of general and/or specific purpose hardware and/or program instructions. In each implementation, the program instructions and hardware can be created using standard programming and engineering techniques.
(29) FIG. **4** shows a functional block diagram illustrating an imaging controller **105** in accordance with aspects of the present disclosure. The imaging controller **105** can be the same or similar to that previously described herein. The imaging controller **105** can include a processor **405**, a memory device **407**, a storage device **409**, a network interface **413**, an image processor **421**, an I/O processor **425**, and a data bus **431**. Also, the imaging controller **105** can include image input connections **461**A, **461**B, **461**C, image output connection **463** that receive and transmit image signals from the image processor **421**. Further, the imaging controller **105** can include input/output connections **469**A, **439**B that receive/transmit data signals from I/O processor **425**.
(30) In implementations, the imaging controller **105** can include one or more microprocessors, microchips, or application-specific integrated circuits. The memory device **407** can include one or more types of random-access memory (RAM), read-only memory (ROM) and cache memory employed during execution of program instructions. Additionally, the imaging controller **105** can include one or more data buses **431** by which it communicates with the memory device **407**, the storage device **409**, the network interface **413**, the image processor **421**, and the I/O processor **425**. The storage device **409** can comprise a computer-readable, non-volatile hardware storage device that stores information and program instructions. For example, the storage device **409** can be one or more, flash drives and/or hard disk drives.
(31) The I/O processor **425** can be connected the processor **405** can include any device that enables an individual to interact with the processor **405** (e.g., a user interface) and/or any device that enables the processor **405** to communicate with one or more other computing devices using any type of communications link. The I/O processor **425** can generate and receive, for example, digital and analog inputs/outputs according to various data transmission protocols.
(32) The processor **405** executes program instructions (e.g., an operating system and/or application programs), which can be stored in the memory device **407** and/or the storage device **409**. The processor **405** can also execute program instructions of an image processing module **455** and an image combination module **459**. The image processing module **455** can be configured to stabilize the images to reduce the blurring, compensate for differences in tilt and rotation, remove reflections

and other visual artifacts from the images, and normalize the images. Additionally, the image processing module **455** can be configured to identify and characterize structures, such as tools or tissues, in the images. Further, the imaging processing module **455** can be configured to determine obstructions in the overlapping fields of view and process the images streams **127**A, **127**B to remove the obstructions.

(33) The image combination module **459** can be configured to analyze images received in an image streams **127**A, **127**B from the imaging units and combine them into a single, combined image stream **133** based on the spatial information. In some implementations, the image combination module **459** generates the combination image stream **133** by registering and stitching together the image streams **127**A, **127**B based on the respective fields-of-view of the imaging units. In some implementations, either of the imaging units can be selected by an operator as a primary imaging unit (e.g., imaging unit **111**A), and the image combination module **459** can generate the combination image stream **133** by using the image stream **127**B of the secondary imaging unit to augment the image stream **127**A. The combination image stream **133** can provide an improved field of view than the field of view of the image stream **127**A. The combination image stream **133** can also provide a 3D view from the perspective of the primary imaging unit. In some implementations, the combination image stream **133** lacks the obstructions removed by the image processing module **455**. In some implementations, the combination image stream **133** also includes images provided by a secondary imaging system (e.g., secondary imaging system **115**).

(34) It is noted that the imaging controller **105** is only representative of various possible equivalent-computing devices that can perform the processes and functions described herein. To this extent, in some implementations, the functionality provided by the imaging controller **105** can be any combination of general and/or specific purpose hardware and/or program instructions. In each implementation, the program instructions and hardware can be created using standard programming and engineering techniques.

(35) The flow diagram in FIGS. **5**A and **5**B illustrates an example of the functionality and operation of possible implementations of systems, methods, and computer program products according to various implementations consistent with the present disclosure. Each block in the flow diagram of FIGS. **5**A and **5**B can represent a module, segment, or portion of program instructions, which includes one or more computer executable instructions for implementing the illustrated functions and operations. In some alternative implementations, the functions and/or operations illustrated in a particular block of the flow diagram can occur out of the order shown in FIGS. **5**A and **5**B. For example, two blocks shown in succession can be executed substantially concurrently, or the blocks can sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the flow diagram and combinations of blocks in the block can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(36) FIGS. **5**A and **5**B show a flow block diagram illustrating an example of a method **500** for generating a combined image stream from individual image streams generated by multiple different imaging units. Referring to FIG. **5**A, at block **505** the system receives image streams (e.g., image streams **127**A, **127**B) from a first imaging unit (e.g., imaging unit **111**A) and a second imaging unit (imaging unit **111**B). For example, the imaging units can be included in a cannula placed within a subject's (e.g., subject **117**) abdomen by a surgeon.

(37) At block **509**, the system can receive spatial information (e.g., spatial information **129**A, **129**B) from the first imaging unit and the second imaging unit. The spatial information can be time-synchronized with the respective image streams received at block **505**. In some implementations, the spatial information is received substantially simultaneously and substantially synchronously with the image streams received at block **505**. The spatial information can include distances, angles, and rotations of the imaging units with respect to one another or with respect to a reference position (e.g., top dead center).

(38) At block **513**, the system can receive images (e.g., alternate image stream **131**) from a secondary imaging system (e.g., secondary imaging system **115**), which can be a CT scanner, X-ray imager, an ultrasound system, or a fluorescence imaging system, for example. In some implementations, the images from the secondary imaging system can received substantially simultaneously and substantially synchronously with the respective image streams received at block **505** and the spatial information received at block **509**.

(39) At block **517**, the system can determine spatial relationships between the first imaging unit and the second imaging unit. In some implementations, determining the spatial relationships includes determining overlapping fields-of view-of the imaging units. Additionally, in some implementations, determining the spatial relationships includes determining a relative distance, angle, and orientation of the imaging units. For example, using the spatial information, the system can determine the relative distance, angle, and rotation of the sensor housings. Further, in some implementations, determining the spatial relationships between the first imaging unit and the second imaging unit can include analyzing respective images of the first imaging unit and the second imaging unit using computer vision techniques.

(40) At block **521**, the system determines whether the spatial relationship between determined at block **517** exceeds one or more predefined thresholds. For example, the system can determine whether an angle (e.g., angle **137**) and rotation of first imaging unit relative to the second imaging unit exceeds respective maximum values (e.g. 45 degrees). If the system determines that one or more of the spatial relationships exceeds their thresholds at block **521** (e.g., block **521** is “Yes”), then at block **523** the system can issue a notification indicating to an operator to reposition the imaging units to adjust their alignment and the method can iteratively return to block **505**. In some implementations, the notification can be displayed (e.g., via screen **145** using display **107**) to the operator along with guidance for correcting the issue. The guidance can include, for example, positive and negative feedback indicating positional, angular, and rotational changes to bring spatial relationship within the maximum values.

(41) On the other hand, if the system determines at block **521** that the distance, angle, and orientation is less than or equal to a respective one of the thresholds, then as indicated by off-page connector “A,” at block **525** of FIG. **5**B, the system (e.g., executing image processing module **455**) can process the image streams received at block **505** to correct the images received from the first and second imaging units. Correcting the images can include stabilizing the images to reduce the blurring of the images due to of vibrations and other movements of the imaging units, individually and with respect to one another. Correcting the images can also include correcting the spatial relationships of the images to compensate for differences in tilt and rotation of the imaging units. Further, correcting the images can include removing specular reflections and other visual artifacts from the images. Moreover, correcting the images can include normalizing the images for exposure, color, contrast, and the like.

(42) At block **531**, the system can identify and characterize structures in images from first imaging unit and second imaging unit. In some implementations, the system can analyze the images of the individual or the combined image streams to identify regions and boundaries of particular structures. For example, the system can extract features to obtain a collection of locations and data, and interpolated it using pattern recognition algorithms for example, to determine the shapes and relative spatial positions of structures. At block **535**, the system can remove obstructions between the first imaging unit and the second imaging unit based on the spatial relationships determined at block **517** and the structures identified at block **531**. For example, the system remove the sensor housing or tissues from images comprising the combined image stream.

(43) At block **539**, the system can generate a first combined image stream (e.g., combination image stream **133**) using the image streams received at block **505** or modified at blocks **525** and **535** using the spatial information received at block **509** and the spatial relationships determined at block **517**. For example, as described above, the system can register and stitch together images in the image

streams to provide an augmented field-of-view. At block **541**, the system can generate the combined image stream by also combining the image stream generated at block **539** with the images received in an image stream from the secondary source at block **513**. At block **543**, the system can generate a display (e.g., an images **145**) using the image streams combined at blocks **527** or **531** on a display device (e.g., display device **107**). The method can then iteratively return to block **505** of FIG. **5**A, as indicated by off-page connector "B."

(44) The present disclosure is not to be limited in terms of the particular implementations described in this application, which are intended as illustrations of various aspects. Many modifications and variations can be made without departing from its spirit and scope, as will be apparent to those skilled in the art. Functionally equivalent methods and apparatuses within the scope of the disclosure, in addition to those enumerated herein, will be apparent to those skilled in the art from the foregoing descriptions. Such modifications and variations are intended to fall within the scope of the appended claims. The present disclosure is to be limited only by the terms of the appended claims, along with the full scope of equivalents to which such claims are entitled. It is also to be understood that the terminology used herein is for the purpose of describing examples of implementations, and is not intended to be limiting.

(45) With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations may be expressly set forth herein for sake of clarity.

(46) It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as "open" terms (e.g., the term "including" should be interpreted as "including but not limited to," the term "having" should be interpreted as "having at least," the term "includes" should be interpreted as "includes but is not limited to," etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases "at least one" and "one or more" to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles "a" or "an" limits any particular claim containing such introduced claim recitation to some implementations containing only one such recitation, even when the same claim includes the introductory phrases "one or more" or "at least one" and indefinite articles such as "a" or "an" (e.g., "a" and/or "an" should be interpreted to mean "at least one" or "one or more"); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should be interpreted to mean at least the recited number (e.g., the bare recitation of "two recitations," without other modifiers, means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to "at least one of A, B, and C, etc." is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., "a system having at least one of A, B, and C" would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to "at least one of A, B, or C, etc." is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., "a system having at least one of A, B, or C" would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the

terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.” In addition, where features or aspects of the disclosure are described in terms of Markush groups, those skilled in the art will recognize that the disclosure is also thereby described in terms of any individual member or subgroup of members of the Markush group.

Claims

1. An imaging system comprising: a display device; an imaging controller; a first imaging unit configured to be inserted into a body of a subject; a second imaging unit configured to be inserted into a the body of the subject; and a processor, wherein the imaging controller includes a computer-readable data storage device storing program instructions that, when executed by the processor, control the system to: receive a first image stream and spatial information from the first imaging unit; receive a second image stream from the second imaging unit; combine the first image stream and the second image stream based on the spatial information; determine spatial relationships between the first imaging unit and the second imaging unit; determine whether the spatial relationships exceed or are below one or more thresholds; and in response to determining whether the spatial relationships exceed or are below the one or more thresholds, provide a notification to an operator or process the first and second image streams to correct images received.
2. The imaging system of claim 1, wherein the spatial information comprises distance, rotation, and angle information of the first imaging unit relative to the second imaging unit.
3. The imaging system of claim 1, wherein the first imaging unit and the second imaging units individually comprise a cannula and a sensor housing.
4. The imaging system of claim 3, wherein: the sensor housing of the first imaging unit comprises a first imaging sensor configured to generate the first image stream; and the sensor housing of the second imaging unit comprises a second imaging sensor configure to generate the second image stream.
5. The imaging system of claim 4, wherein: the sensor housing of the first imaging unit comprises a first LiDAR device configured to generate the spatial information.
6. The imaging system of claim 5, wherein the spatial information further comprises a distance and rotation of the first imaging unit relative to the second imaging unit.
7. The imaging system of claim 1, wherein combining the first image stream and the second image stream comprises registering and stitching images of the first image stream with images of the second image stream.
8. The imaging system of claim 7, wherein the images of the first image stream, the images of the second image stream, and the spatial information are time- synchronized.
9. The imaging system of claim 1, wherein program instructions further control the system to: display, using the combination of the first image stream and the second image stream, an augmented version of the second image stream.
10. The endoscopic imaging unit of claim 2, wherein the first imaging unit comprises: a plurality of antennas configured to receive a ranging signal from the second imaging unit; and the spatial information is determined by determining the distances from the second imaging unit based on a difference of arrival times of the ranging signal at the plurality of antennas.
11. The endoscopic imaging unit of claim 3, wherein the cannula is configured to receive an obturator.
12. The imaging system of claim 3, wherein the sensor housing of the first imaging unit comprises one or more image sensors, a LiDAR device, and a light source.
13. The imaging system of claim 10, wherein the plurality of antennas are placed in a substantially straight line on one or more sides of the first and second imaging units.
14. The imaging system of claim 1, wherein the imaging controller is configured to identify and

remove images of physical structures in overlapping fields-of-view.

15. The imaging system of claim 14, wherein the controller is configured to remove the images of physical structures using the first and second image streams and the spatial information.

16. The imaging system of claim 1, wherein the imaging controller provides feedback to an operator via a display device to one or both of translate or rotate one or both of the first and second imaging units to provide overlapping fields-of-view within thresholds.

17. The imaging system of claim 1, wherein the notification is provided in response to the determining that the spatial relationships exceed the one or more thresholds, and wherein the notification in particular comprises positive and negative feedback indicating positional, angular, and rotational changes to bring the spatial relationships within maximum values.

18. The imaging system of claim 1, wherein the spatial relationships comprise a relative distance, angle, and orientation of the imaging units, and wherein the program instructions further control the system to perform the correcting of the images of the first and second image streams in response to the determining that the spatial relationships are less than or equal to the one or more thresholds, wherein the correcting comprises one or more of the following: a) stabilizing the images to reduce blurring; b) correcting the spatial relationships by compensating differences in tilt and rotation of the first and second imaging units; c) removing specular reflections and/or visual artifacts from the images; d) normalizing the images for exposure, color, and/or contrast.

19. The imaging system of claim 1, wherein the maximum values comprise a maximum value for an angle of the first imaging unit relative to the second imaging unit wherein the maximum value for the angle is 45 degrees.